



Risk Analysis and Assessment

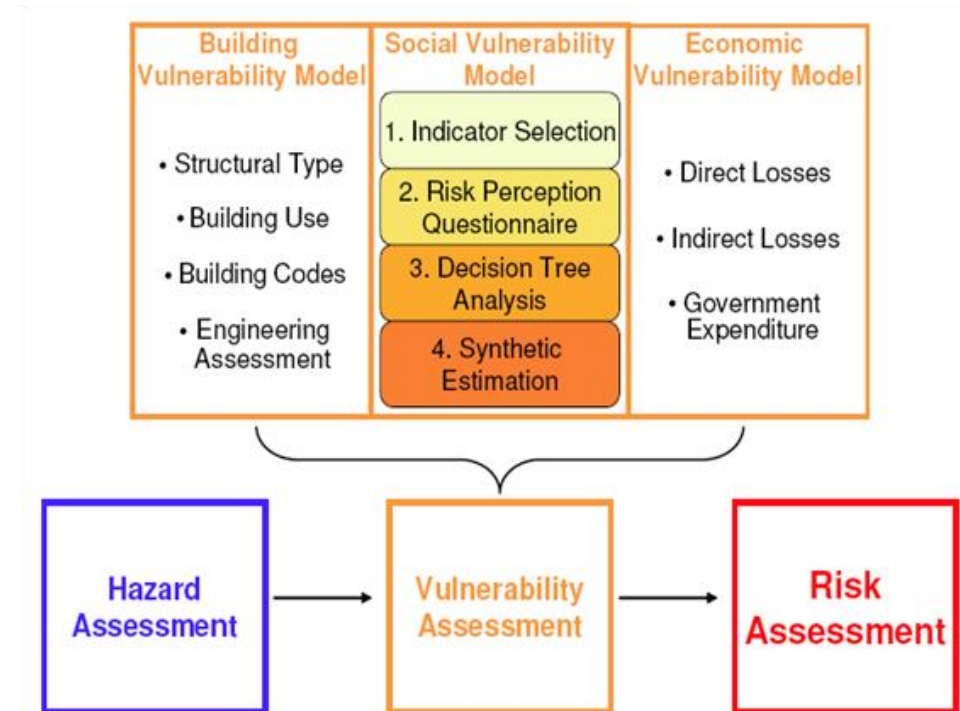
Fall 2026

A risk matrix

The risk matrix is structured as follows:

PROBABILITY		POTENTIAL LOSS			
		Low	Medium	High	Very High
High	C	B	A	A	
Medium	C	B	B	A	
Low	D	C	B	B	
Very low	D	D	C	C	

The matrix uses a color-coded system to represent risk levels: Red for 'A' (High Risk), Orange for 'B' (Medium Risk), Yellow for 'C' (Low Risk), and Green for 'D' (Very Low Risk).





Learning Outcome

Students will Learn how to analyze and assess disaster related risks

Reading Material

- Dr. Cees Westen (ITC) Guide Book 01 (pdf provided)

Recap - Risk?

- Probability that loss will occur as the result of an adverse event, given the hazard and the vulnerability
- What is at Risk? - consequence of Hazards
 - Personal Injury
 - Destruction of property
 - Poor building material and maintenance
 - Damage to local economy
 - Opportunity Loss:
 - diversion of economic resources to recovery efforts

Risks can also be prioritized according to the acceptability of the population at risk.

How?

Risk and Risk Assessment

Definition of risk

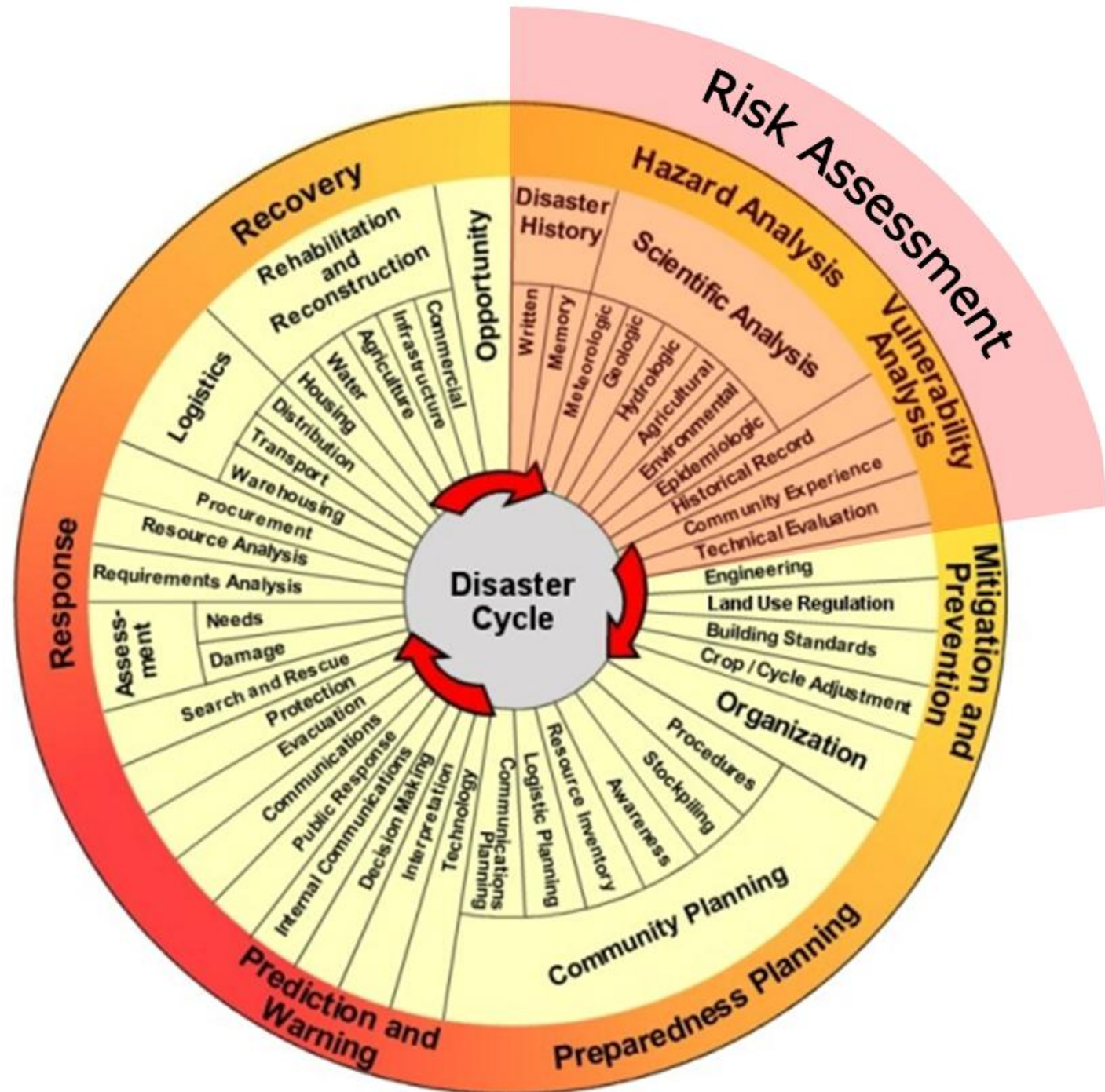
- the probability of harmful consequences, or expected losses (deaths, injuries, property, livelihoods, economic activity disrupted or environment damaged) resulting from interactions between (natural, human-induced or man-made) hazards and vulnerable conditions

Definition of risk assessment:

- A methodology to determine the nature and extent of risk by analyzing potential hazards and evaluating existing conditions of vulnerability that could pose a potential threat or harm to people, livelihoods and the environment on which they depend.

(Source UN-ISDR: <http://www.unisdr.org/eng/library/lib-terminology-eng%20home.htm>)

The process of conducting a risk assessment is based on a review of both technical features of hazards such as their location, intensity, frequency/probability and also the analysis of the physical, social, economic and environmental dimensions of vulnerability and exposure, while taking into account of the coping capacities pertinent to the risk



Risk = Probability of Losses

$\text{Risk} = \text{Hazard} * \text{Vulnerability} * \text{Amount of elements-at-risk}$

Or

$\text{Risk} = \text{Hazard} * \text{Vulnerability} / \text{Capacity}$

- **Capacity** = Expresses the positive managerial and operational resources and procedures for reducing risk factors

Examples:

- awareness raising
- early warning
- preparedness planning

Hazard Risk Assessment

- Process by which **threat** posed by each identified hazard is investigated.
- Risk depends on Hazard likelihood and hazard consequences.
- The above two (02) factors are constantly changing as a result of changing environment.
- High-likelihood/high-consequence hazards are of special consideration.
- A comprehensive Risk Management Program can greatly reduce life and property losses.

$$\text{RISK} = \text{HAZARD} * \text{VULNERABILITY} * \text{AMOUNT}$$

- **Hazard** = Probability of an event with a certain magnitude
 - Triggering mechanism
 - Type of event
 - Magnitude: Volume, distance, duration, speed, etc
- **Vulnerability** = Degree of damage. Function of:
 - Magnitude of event, and
 - Type of elements at risk
- **Amount** = Quantification of the elements at risk, e.g.
 - Replacement costs of buildings, infrastructure, etc.
 - Loss of function or economic activities
 - Number of people

Vulnerability of five (05) elements of a community

- Each of those elements has a specific vulnerability to a specific hazard



Types of Vulnerability	Characteristics
Tangible/Material (easy to see; value easily determined)	People – lives, health, security, living conditions Property – services, physical property loss, loss of use Economy – loss of products and production, income Environment – water, soil, air, vegetation, wildlife
Intangible/Abstract (difficult to see; value difficult to determine)	Social structures – family and community relationships Cultural practices – religious and agricultural Cohesion – disruption of normal life Motivation – will to recover; government response

Vulnerability Assessment

- The process of estimating the vulnerability to potential disaster hazards of specified elements at risk.
- A process to determine the nature and extent of vulnerability by analyzing existing conditions and characteristics of an element or system exposed to a particular type of natural hazard event.

Vulnerability Assessment

- Vulnerability assessment identifies the relationship between the hazard and the element at risk.
- Vulnerability is evaluated as the interaction between the intensity of the hazard and the type of element at-risk, making use of so-called *vulnerability curves*

How to do Vulnerability Assessment?

- Which are the most important elements at risk?
- Which methods for vulnerability assessment could be used?
- What are the data requirements?
- Aggravating factors of vulnerability? Examples?

Risk Assessment

- For different hazards there are different
 - Methods for loss estimation
 - Data requirements

Risk Assessment = Risk Estimation + Risk Evaluation

Risk Analysis Methodology

- Both Qualitative and Quantitative.
- Various computer based models.
- Risk assessment outcomes depend on quality and availability of data.

Qualitative Risk Analysis

- An analysis which uses word form, descriptive or numeric rating scales to describe the magnitude of potential consequences and the likelihood that those consequences will occur
- Resulting in a qualitative value of the risk. e.g. High Risk, Moderate Risk, Low Risk etc.

Quantitative Risk Analysis

- An analysis which is based on numerical values of the probability, vulnerability, and consequences
- Resulting in a numerical value of the risk. e.g.

Risk = \$ 5,000, etc.

Risk matrix



Probability		Loss
High	Events that occur more frequently than once in 10 years	Based on the potential by taking into account elements at risk Fatalities Injuries Impact on facilities, critical services and infrastructure Property damage Business interruption Environmental/ Economic impact
Moderate	Events that occur from once in 10 years to once in 100 years	
Low	Events that occur from once in 100 years to once in 1000 years	
Very Low	Events that occur less frequently than once in 1000 years	

The risk matrix



A risk matrix					
↑ P R O B A B I L I T Y	High	C	B	A	A
	Medium	C	B	B	A
	Low	D	C	B	B
	Very low	D	D	C	C
		Low	Medium	High	Very High
					→ P O T E N T I A L L O S S

Risk matrix



Class A

High-risk condition. Immediate action is necessary.

Possible deaths over 1000

People affected may be over 100,000

Complete shut down of facilities and critical services for more than 14 days

Over 50% of property located in the area may be damaged

Class B

Moderate to high-risk condition where risk can be reduced by mitigation activities and contingency planning. Prompt attention needed.

Possible deaths less than 1000

People affected may be between 50,000 to 100,000

Complete shut down of facilities and critical services for 7 days

25% of the property located in the area may be damaged

Class C

Low risk condition. However investment in further mitigation and planning may be necessary after cost-benefit analysis.

Possibility of death low.

People affected between 10,000 and 50,000

Complete shut down of facilities not more than 1 day

About 10% of the property located in the area may be damaged

Class D

Very low risk condition. Only limited action necessary.

No possibility of death

People affected less than 10,000

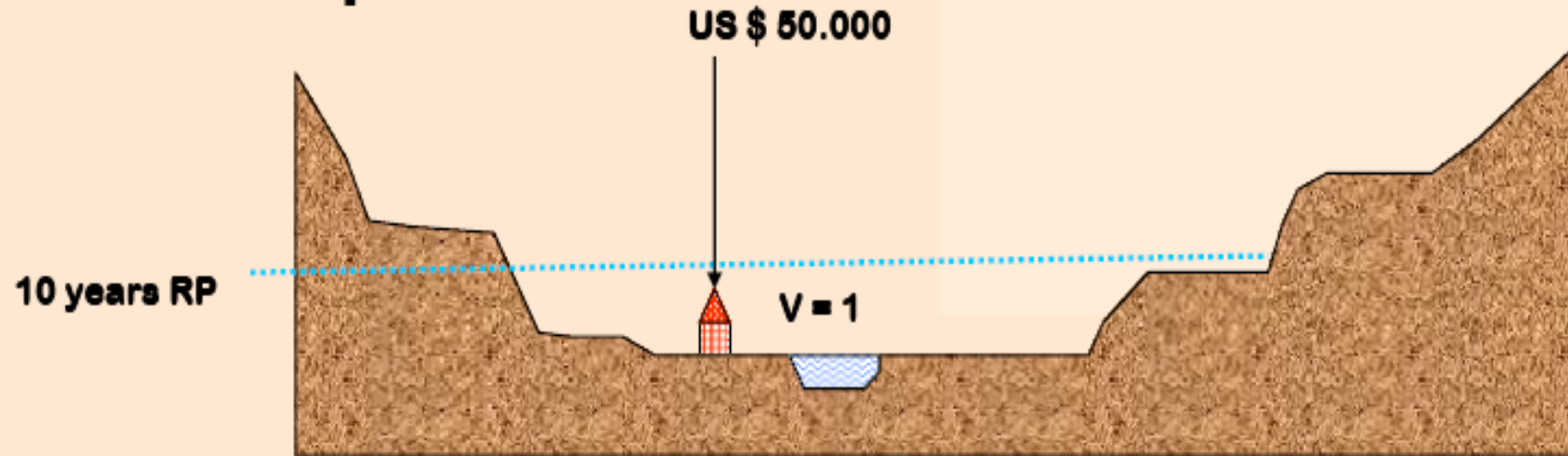
Facilities and critical services may not be affected

About 1% of the property located in the area may be damaged

Hazard, vulnerability and risk?



- **Example:**

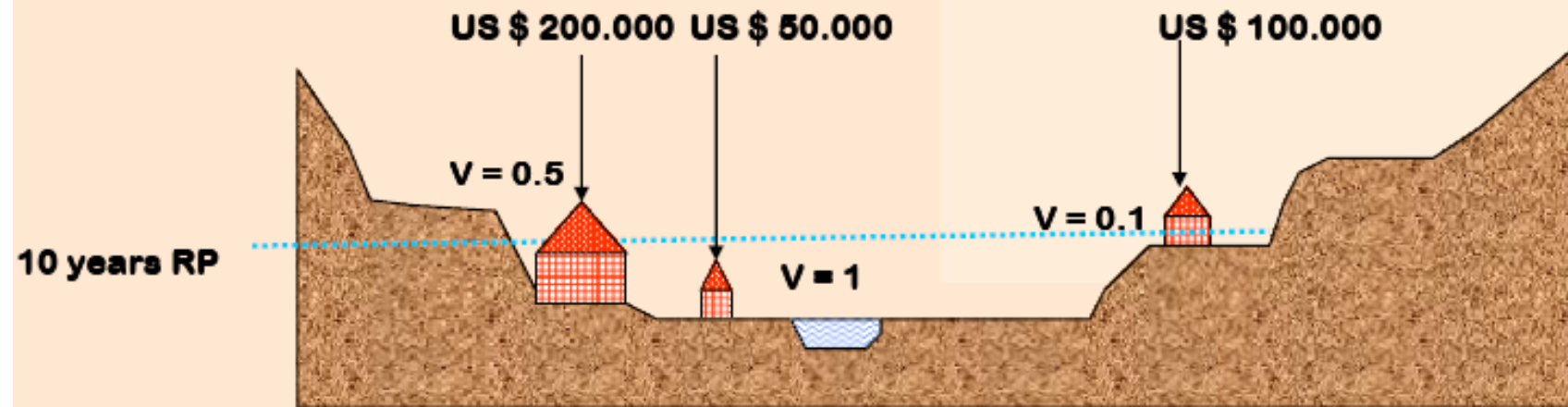


- Hazard = probability within a given period
= 0.1 / year
- Risk = hazard * vulnerability * amount
= 0.1 * 1 * 50.000 = 5.000 US \$

Hazard, vulnerability and risk?



• Example:



- Risk

$$= \text{hazard} * \text{vulnerability} * \text{amount}$$

$$= 0.1 * ($$

$$(0.5 * 200.000) +$$

$$(0.1 * 100.000) +$$

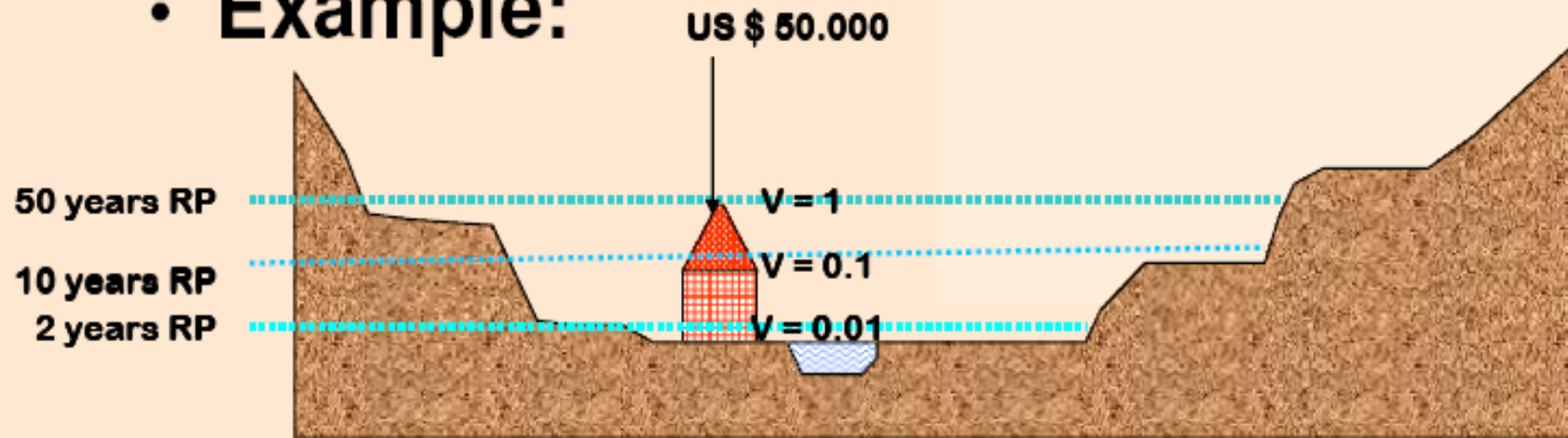
$$(1 * 50.000))$$

$$= 0.1 * 160.000 = 16.000 \$$$

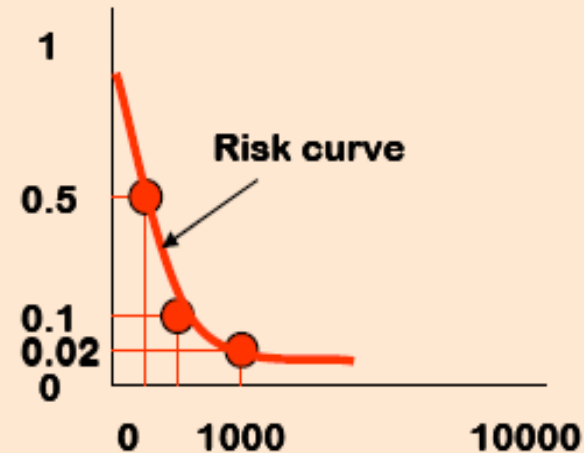
Hazard, vulnerability and risk?



• Example:



- $$\begin{aligned} \text{Hazard} &= 0.5 * 0.01 * 50.000 + \\ & 0.1 * 0.1 * 50.000 + \\ & 0.02 * 1 * 50.000 = \\ & = 250 + 500 + 1000 \\ & = 1750 \text{ US \$} \end{aligned}$$



The risk curve

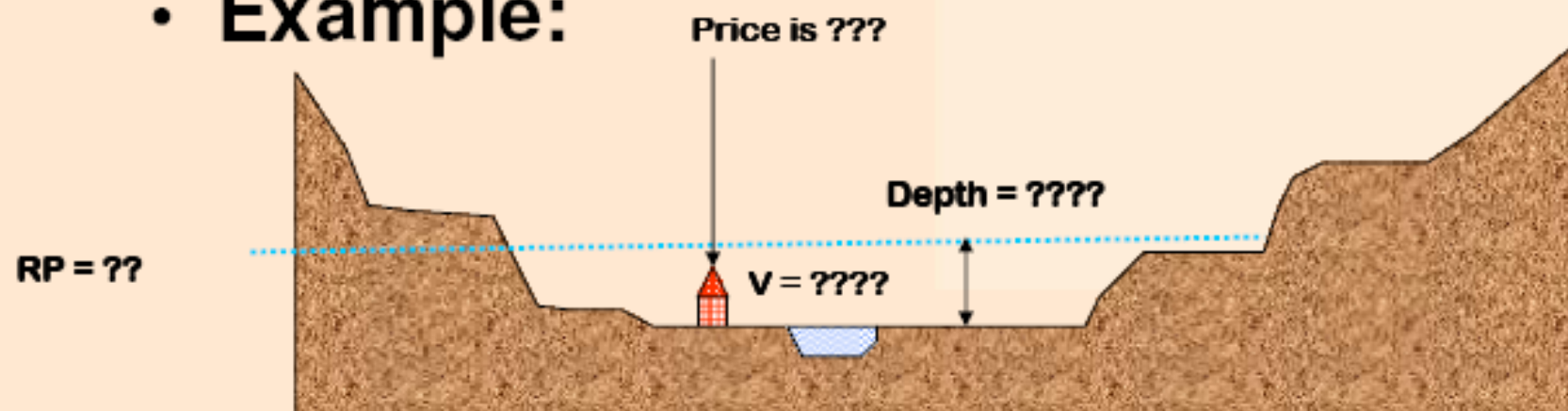


- a graphic representation of the probability of a specific loss associated with a specific return period, such as that expected once every 500 years, 100 years, 50 years, etc
- a complete reporting of potential loss from all possible outcomes from a full spectrum of possible events
- For example,
- a 1% probability of exceedance (1% chance of being larger) is the loss level with a 99% chance of not being exceeded annually, which translates into a 100-year loss return period. Likewise, 0.2% equates to a 500 return period loss

In reality



- Example:**



- Risk = hazard * vulnerability * amount
 = ? * ? * ? = unknown

What is needed:

hazard assessment , elements at risk
 mapping, vulnerability assessment, cost
 estimation.

Information required



- **For hazard assessment**
 - Scientific studies/maps, long-term monitoring, historic
 - reports on past incidence of hazards, in particular the
 - location, frequency and severity of the events
 - Understanding of factors influencing hazard occurrence
- **For vulnerability assessment**
 - Hazard characteristics (from the hazard assessment)
 - Characteristics of the element/system of interest:
 - location/site, structural strength/quality, content exposure
 - Environmental quality
- **For risk assessment**
 - Same as for vulnerability assessment, plus specific damage
 - / loss frequency estimates for elements at risk
 - Specific hazard scenario

Risk reduction. Example



In 1977, a cyclone struck the Andhra Pradesh. The following loss figures were estimated for the East Godavari District.

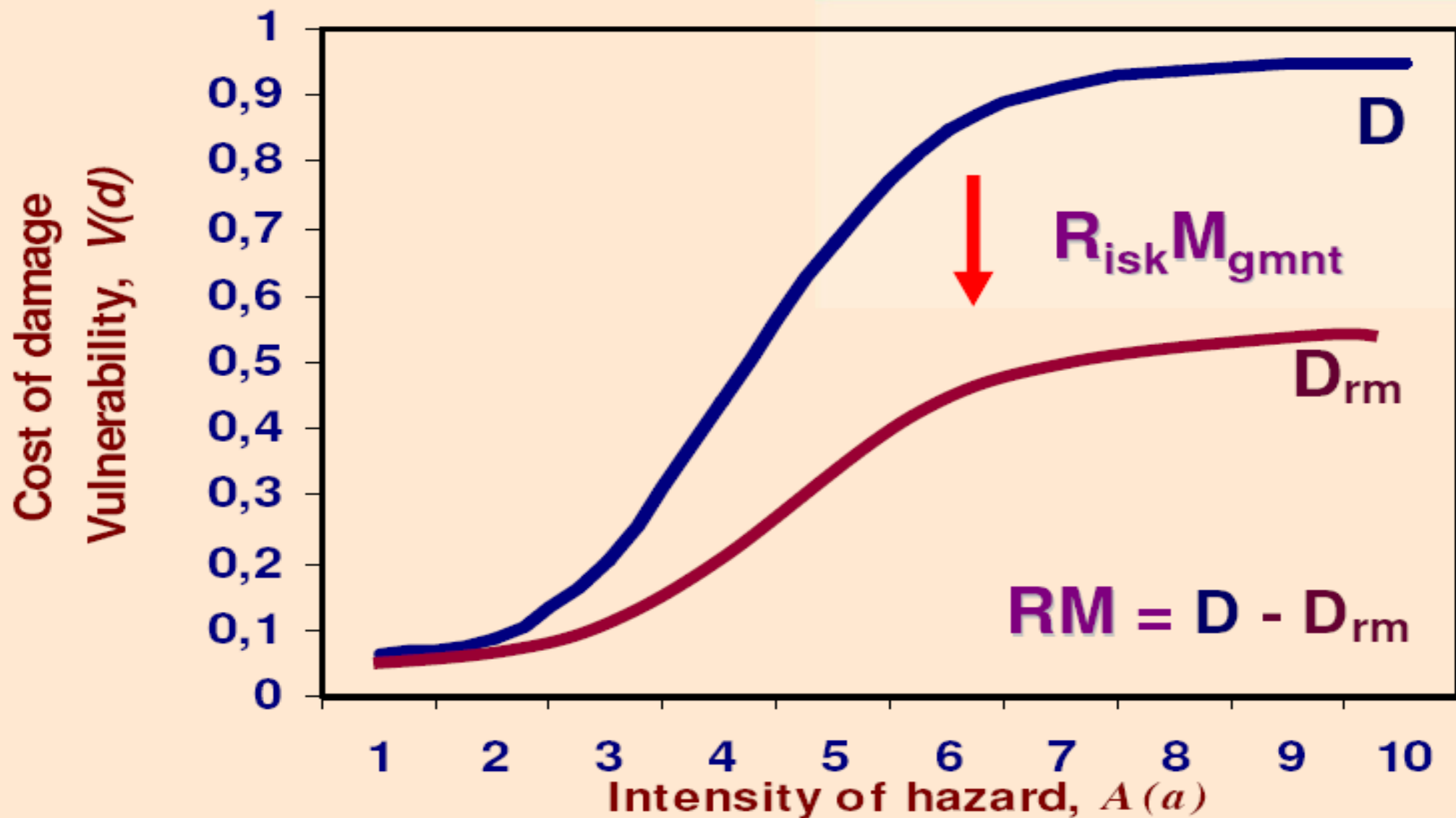
Houses destroyed	2,89,906
Houses damaged	89,677
TOTAL	3,79,583
Persons rendered homeless	1.442 million

If measures were taken to strengthen houses by retrofitting, the loss would have been reduced as follows.

Houses destroyed	Nil
Houses damaged	1,36,489
Persons rendered homeless	0.519 million

The economic benefit is estimated to be Rs. 91 crores.

Reducing the cost of damage through risk management



Elements at Risk

- All objects, persons, animals, activities and processes that may be adversely affected by hazardous phenomena, in a particular area, either directly or indirectly.
- This includes: buildings, facilities, population, livestock, economic activities, public services, environment.

Elements at risk



Elements at risk may be listed as follows:

- **Physical**

Infrastructure

Roads

Railway

Bridges

Harbour

Airport

Critical facilities

Emergency shelters

Schools

Hospitals and Nursing Homes

Fire Brigades

Police

Utilities

Power supply

Water supply

Transport

Communication

Government services

- **Economic**

Business and trade activities

Access to work

Impact on work force

Opportunity cost

- **Societal**

Vulnerable age categories

Low income group people

Gender

- **Environmental**

Loss of biodiversity

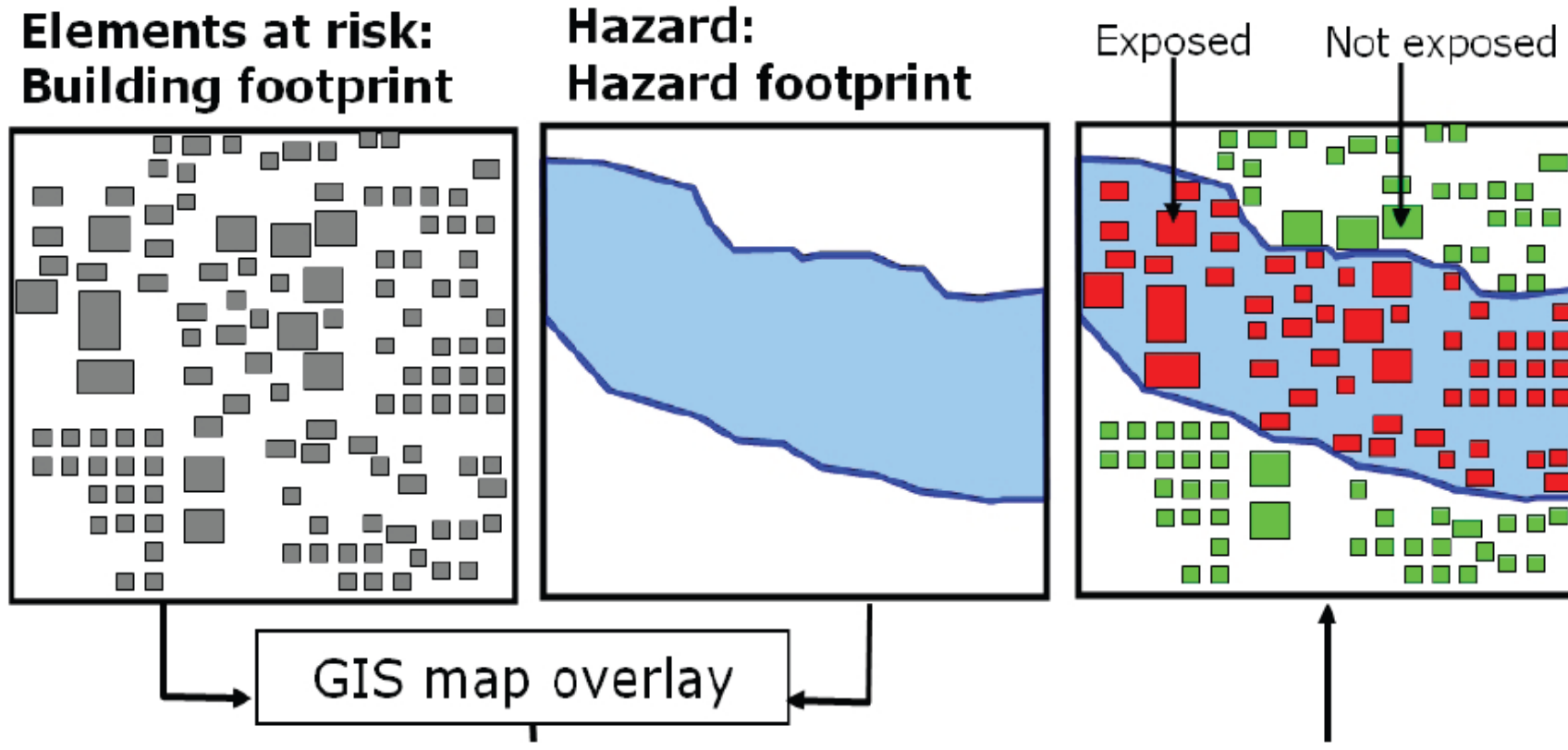
Damaged landscape

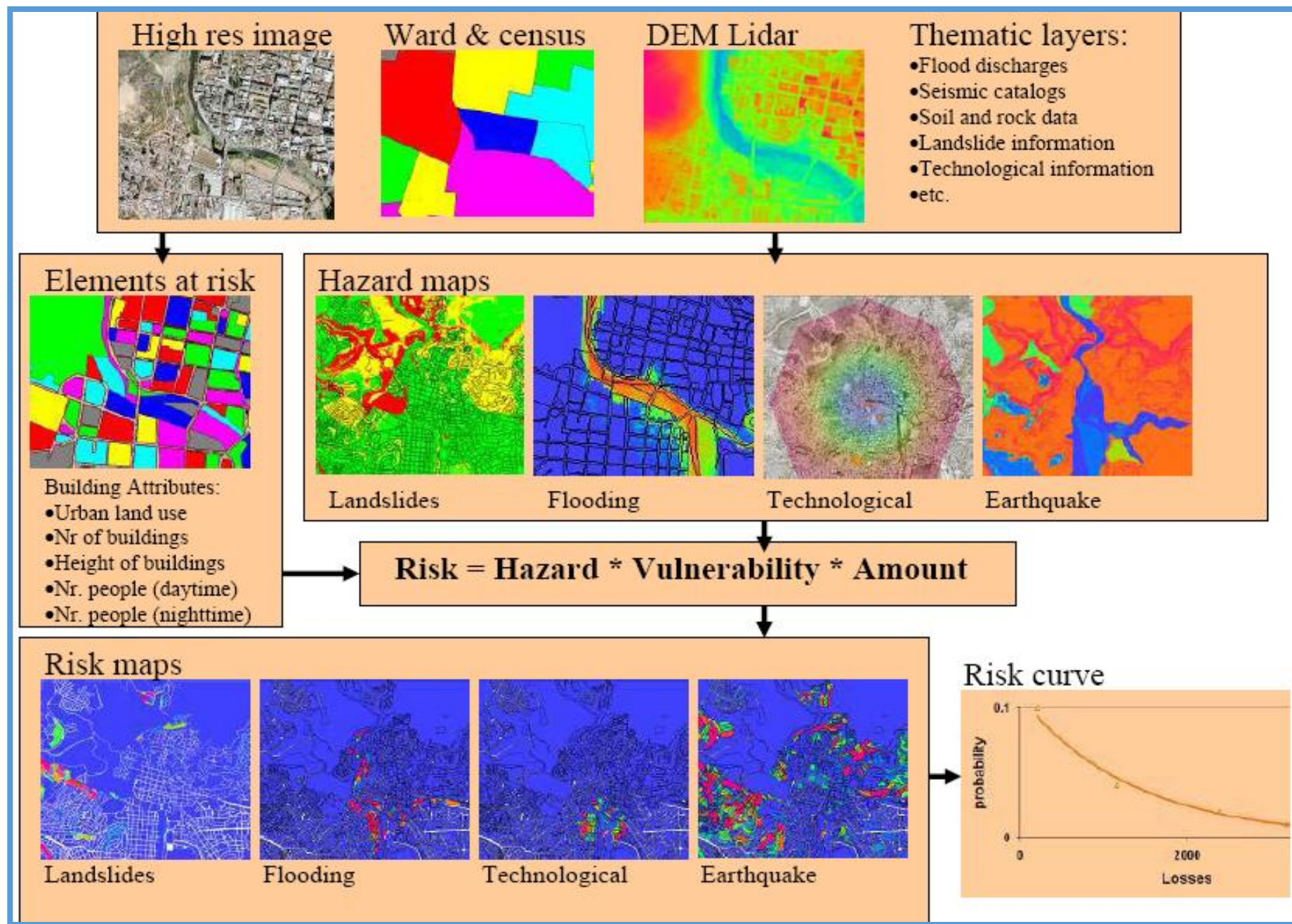
Physical and chemical changes
in the surroundings

Elements of Risk Matrix

Hazard Type	Element at Risk		Sources (bold = primary source)			
			Register	Map	Field	Image
Earthquake	Buildings	Building type	√	√	√	√
		Structure type				
		Material				
		Height				
		Use				
		Shape				
		age				
	Lifelines					
	Population					
	Essential facilities					
Flood						

Exposure to Hazard





Summary

- Three major components to be considered in disaster management
 - Hazard
 - Vulnerability
 - Capacity/amount of losses
- Therefore, disaster risk can be reduced through systematic efforts of
 - Reducing hazard exposure,
 - Lessening vulnerability and/or
 - Developing a greater capacity to face these disasters may reduce the impact of a hazard

Discussion/Comment/Question

